

Chi-Sheng (Michael) Chen (陳麒升)

AI in Medicine/Finance | EEG/BCI | Multi-Modal Learning | Quantum Machine Learning

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Nationality: Taiwan | Country of normal residence (last 5 years): Taiwan and USA | Based in Cambridge, MA, USA

Chi-Sheng (Michael) Chen is a biomedical AI researcher specializing in classical/quantum ML for clinical biosignals and foundation models, with 40+ publications (JAD IF=6.5, Physical Review A, ICASSP, QCE), 320+ citations, and reviewer service at TPAMI, NSR, and npj Quantum Information. His EEG models are clinically deployed at Taipei Veterans General Hospital, and his time-series models run in production for DeFi AMM strategies.

RESEARCH & INDUSTRY EXPERIENCE

Harvard Medical School & Beth Israel Deaconess Medical Center

Research Affiliate, Advisor: Prof. Dr. Gabriel Brat

MA, USA

Nov 2024 – Now

- Developing real-time EMS triage pipeline using multimodal AI for trauma prediction.
- Collaborating with surgeons on AI-assisted decision support systems.

OpenAI Inc.

AI Trainer (Contractor)

MA, USA

Mar 2025 – Oct 2025

- Performed high-complexity AI data labeling and evaluation tasks, including instruction following, multimodal reasoning, and safety alignment.
- Contributed to Reinforcement Learning from Human Feedback (RLHF) pipelines by providing high-quality comparative judgments, preference rankings, and model-behavior assessments.

Omnis Labs

Co-Founder

Remote

Sep 2024 – Now

- AI-driven DeFi liquidity positioning protocol with automated market maker (AMM) strategies and backtesting systems.
- Achieved 50%–120% base fee APR on WBTC/USDC and ETH/USDC pairs (Katana on SushiSwap) using time-series model-driven AMM strategies.

Neuro Industry, Inc.

Researcher, Co-Founder & CTO

CA, USA

Mar 2024 – Jan 2025

- Built GCP-based MLOps pipeline, enabling training on 10K+ EEG samples.
- 2 research papers on quantum machine learning on EEG signal processing.
- 1 research paper on EEG-to-image using diffusion model.

Department of Computer Science, National Yang Ming Chiao Tung University

Research Assistant, Advisor: Prof. Chun-Shu Wei

Hsinchu, Taiwan

Dec 2023 – Aug 2024

- Developed MUSE, a multimodal contrastive learning framework for EEG-based image recognition (22 citations).
- Research on nonstationary EEG dynamics and foundation models for brain-computer interfaces.

MediaTek Inc.

Digital IC R&D Engineer

Hsinchu, Taiwan

Dec 2021 – Oct 2023

- Responsible for microprocessor IP development for flagship 5G smartphones' display and AMBA SoC implement.
- Hardware virtualization architecture RTL design and IP verification with UVM and SystemVerilog.

Department of Surgery, National Taiwan University Hospital

Research Assistant, Advisor: Dr. Shuo-Lun Lai

Taipei, Taiwan

Jul 2021 – Sep 2021

- Deep learning applications on surgery automation with YOLO-based model.
- Developed a prototype for a surgical smoke evacuation system.

Department of Psychiatry, Taipei Veterans General Hospital

Research Assistant, Advisor: Prof. Dr. Cheng-Ta Li

Taipei, Taiwan

Sep 2019 – Jun 2021

- Collaborated with researchers and clinicians to identify patterns and trends in brainwave activity related to major depressive disorder.
- Utilized machine learning algorithms to develop predictive models based on real-world brainwave data.

Max Planck Institute for Chemical Physics of Solids (MPI CPfS)

Research Internship, Advisor: Dr. Alexander Komarek & Dr. Li Zhao

Dresden, Germany

Jul 2018 – Sep 2018

- Searching new possible unconventional superconductors among Co-based quaternary chalcogenides with diamond like structure $\text{CuInCo}_2\text{A}_4$ / $\text{AgInCo}_2\text{A}_4$ (A = Te, Se, S).

EDUCATION

National Taiwan University (NTU)

Master of Science in Biomedical Electronics and Bioinformatics GPA: 3.74/4.30
Advisor: Prof. Dr. Cheng-Ta Li, MD & Prof. Chung-Ping Chen

Taipei, Taiwan

Sep 2019 – Jun 2021

- Machine learning application on non-stationary time series data.
- The research EEG model for TMS pattern personalized prediction is currently being applied to real outpatient patients in the Precision Depression Intervention Center (PreDIC), Taipei Veterans General Hospital.

National Chiao Tung University (NCTU)

Bachelor of Engineering in Electrophysics
Bachelor of Science in Interdisciplinary Science Degree Program

Hsinchu, Taiwan

Sep 2015 – Jun 2019

- Researching the oxide processing techniques.
- Researching the biomedical signal processing on traditional Chinese medicine data.

SKILLS

Programming: C/C++, JavaScript, Python, Rust, MATLAB, R, MySQL, Verilog, SystemVerilog, Perl, Solidity

Frameworks: PyTorch, HuggingFace, Scikit-learn, MNE, MLflow

Tools: Git, Docker, GCP, AWS, HSPICE, Genus, SG-Lint, Verdi, UVM

Languages: Chinese (Native), English (Fluent)

SELECTED PUBLICATIONS (320+ CITES, H-INDEX 11, I10-INDEX 12)

[Peer-Reviewed - First/Co-first Author]

Chen, C.-S., et al. "Exploring the potential of qeegnet for cross-task and cross-dataset electroencephalography encoding with quantum machine learning" Journal of Signal Processing Systems (2025)

Chen, C.-S., et al. "Unraveling quantum environments: Transformer-assisted learning in Lindblad dynamics" Physical Review A (2025), IF=2.9, <https://doi.org/10.1103/gsxk-45mk> .

Chen, C.-S., et al. "Exploring the Potential of Electroencephalography Signal-Based Image Generation Using Diffusion Models: Integrative Framework Combining Mixed Methods and Multimodal Analysis" JMIR MI (2025), IF=3.8, <https://doi.org/10.2196/72027> .

Chen, C.S., et al. "Improving fine-grained food classification using deep residual learning and selective state space models" PloS one (2025), IF=2.9, <https://doi.org/10.1371/journal.pone.0322695> .

Li, C.T., Chen, C.S., et al. "Prediction of antidepressant responses to non-invasive brain stimulation using frontal electroencephalogram signals: Cross-dataset comparisons and validation" Journal of Affective Disorders (2023), IF=6.533, <https://doi.org/10.1016/j.jad.2023.08.059> .

Lai, S.L., Chen, C.S., et al. "Intraoperative Detection of Surgical Gauze Using Deep Convolutional Neural Network" Ann Biomed Eng 51, 352-362 (2023), IF=4.219, <https://doi.org/10.1007/s10439-022-03033-9> .

Chen, C.S., et al. "Quantum Reinforcement Learning-Guided Diffusion Model for Image Synthesis via Hybrid Quantum-Classical Generative Model Architectures" arXiv:2509.14163, 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).

Chen, C.S., et al. "Quantum and Classical Machine Learning in Decentralized Finance: Comparative Evidence from Multi-Asset Backtesting of Automated Market Makers" arXiv:2510.15903, 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).

Chen, C.S., et al. "Quantum Contrastive Learning Framework" arXiv:2408.13919. 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP49660.2025.10889504 .

Chen, C.S., et al. "QEEGNet: Quantum Machine Learning for Enhanced Electroencephalography Encoding" arXiv:2407.19214. IEEE International Workshop on Signal Processing Systems (SiPS) 2024, doi: 10.1109/SiPS62058.2024.00035 .

Chen, C.S., et al. "Q-DPTS: Quantum Differentially Private Time Series Forecasting via Variational Quantum Circuits" IEEE GLOBECOM Workshop 2025, arXiv:2508.05036.

Chen, C.S., et al. "Quantum Reinforcement Learning Trading Agent for Sector Rotation in the Taiwan Stock Market" IEEE QCE QCRL Workshop 2025, arXiv:2506.20930.

[Peer-Reviewed - Contributing Author]

Lai, S.L., Chao, Y.C., Chen, C.S., et al. "Detection and tracking of a gauze sponge in minimally invasive surgery using a YOLO and R-CNN based model" Medical & Biological Engineering & Computing (2025), IF=2.6, <https://doi.org/10.1007/s11517-025-03471-2> .

Takegami D., et al. "Direct imaging of valence orbitals using hard x-ray photoelectron spectroscopy" Phys. Rev. Res. 4, 033108 (2022), IF=4.3, <https://doi.org/10.1103/PhysRevResearch.4.033108> .

Falke J., et al. "Electronic structure of the metallic oxide ReO₃" Phys. Rev. B 103, 115125 (2021), IF=3.908, <https://doi.org/10.1103/PhysRevB.103.115125> .

[Preprints / Under Review]

Chen, C.S., et al. "A Unified SPD Token Transformer Framework for EEG Classification: Systematic Comparison of Geometric Embeddings" arXiv preprint (2026), arxiv:2601.21521.

Chen, C.S., et al. "FreqLens: Interpretable Frequency Attribution for Time Series Forecasting" arXiv preprint (2026), 2602.08768.

Chen, C.S., et al. "Quantum Adaptive Self-Attention for Quantum Transformer Models" arXiv preprint (2025), arXiv:2504.05336.

Chen, C.S., et al. "Large Cognition Model: Towards Pretrained EEG Foundation Model" arXiv preprint (2025), arXiv:2502.17464.

Chen, C.S., et al. "Mind's Eye: Image Recognition by EEG via Multimodal Similarity-Keeping Contrastive Learning" arXiv preprint (2024), arXiv:2406.16910.

PROFESSIONAL SERVICE

Program Committee: QNRL Workshop @ IEEE WCCI 2026

Conference Reviewer (2026): **NeurIPS, ICML, KDD, MICCAI, ICASSP**

IOP Trusted Reviewer (2026), recognized by the Institute of Physics Publishing for quality and commitment in peer review.

Journal Reviewer (2025–present): **Oxford National Science Review (NSR)**, **IEEE Trans. Pattern Analysis and Machine Intelligence (TPAMI)**, **IEEE Trans. Neural Networks and Learning Systems (TNNLS)**, **IEEE Trans. Neural Systems & Rehabilitation Engineering (TNSRE)**, **IEEE Trans. Systems, Man and Cybernetics (TSMC)**, **IEEE Trans. Consumer Electronics**, **IEEE/ACM Trans. Audio, Speech, and Language Processing (TASLP)**, **IEEE Trans. Cognitive and Developmental Systems (TCDS)**, **IEEE Journal of Biomedical and Health Informatics (JBHI)**, **IEEE Signal Processing Letters (SPL)**, **IEEE Access**, **Tsinghua Science and Technology**, **npj Quantum Information**, **EPJ Quantum Technology**, **IOP Journal of Neural Engineering**, **Springer International Journal of Machine Learning and Cybernetics**, **Springer Cognitive Neurodynamics**, **IOP Biomedical Physics & Engineering Express**, **IOP Engineering Research Express**, **AIMS Quantitative Finance and Economics**, **AIMS Public Health**, **IntechOpen AI**, **Computer Science and Robotics Technology**

AWARDS & ACHIEVEMENTS

2026 NVIDIA GTC Top 8 Finalist: "GPU-Accelerated Chinese Brain-to-Text Interface Powered by NVIDIA H100 80GB."

2026 Berkeley RDI AgentX-AgentBeats: Phase 1 Healthcare Agent 3rd Place (Team MadGAA, 1,300+ teams worldwide).

2023 Certificate of the 20th National Innovation Award, clinical research category.

2019 International Blockchain Olympiad (IBCOL): World Finalist competition with 100+ teams.

2017 Microsoft International Imagine Cup : 2nd place in Taiwan.

2016 International Genetically Engineered Machine Competition (iGEM): World Gold medal, Best Applied Design, Best Part Collection, Best Presentation competition with 300+ university teams.